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Cultivar-dependent transcriptional responses of lychee (**Litchi chinensis**) to **Peronophythora litchii**: from exploratory candidate discovery to a prospectively registered genome-wide analysis with cross-context external evaluation

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3 **ABSTRACT**

4 Litchi downy blight, caused by the oomycete **Peronophythora litchii**, is among the most
5 damaging diseases of lychee (**Litchi chinensis** Sonn.), and public RNA-seq cohorts spanning
6 cultivars, tissues, and infection time points offer a natural resource for identifying cultivar-
7 dependent infection responses. Such data are easily over-interpreted, however, when the same
8 cohort is used both to select candidates and to appear to confirm them. Here we unify two
9 stages of a transcriptomic reanalysis of the lychee–**P. litchii** interaction into a single account.
10 In the first, exploratory stage, we reanalyzed leaf RNA-seq data from the cultivars Guiwei and
11 Yurong1 (GSE201243/PRJNA830488), identified 17 and 117 infection-responsive genes in the
12 two cultivars, and prioritized 18 candidates with defense-associated annotations and apparently
13 enriched stress- and hormone-associated promoter elements. In the second, confirmatory
14 stage, we prospectively registered a discovery and external-evaluation protocol that fixed
15 dataset roles, statistical models, thresholds, and evidence vocabulary before any external
16 outcome was examined. A genome-wide negative-binomial cultivar-by-infection interaction
17 analysis of 19,445 expressed genes produced 262 statistical candidates, of which 206 were
18 retained after uniform mapping and gene-model quality control, 19 remained significant under
19 an independent genome-wide edgeR quasi-likelihood analysis, and 16 passed the complete
20 registered robustness procedure. In the locked cross-context evaluation against an independent
21 pericarp time course contrasting a susceptible and a resistant cultivar (PRJNA450886), two
22 genes were supported and five were directionally contradictory; the internally robust and
23 externally supported sets did not overlap, an outcome compatible with chance given the set
24 sizes. A formal audit showed that none of the original 18 exploratory candidates met the newly
25 applied genome-wide interaction criterion and that two failed uniform-mappability control.
26 Deterministic evidence integration assigned no gene to the highest confidence tier, placed 12
27 internally robust genes, whose annotations include carbohydrate and cell-wall remodeling, lipid-
28 transfer, and chaperone-regulatory functions, in a middle tier, and retired 65 entities, with all null
29 and contradictory outcomes reported. The unified analysis shows that candidates selected for

effect-size prominence in a single cohort largely fail a formally specified genome-wide interaction test, delimits exactly which computational claims about this pathosystem the current public data can support, and releases a candidate resource whose evidence boundaries are explicit. Experimental perturbation remains necessary for any causal conclusion.

****Keywords:**** *Litchi chinensis*; *Peronophythora litchii*; downy blight; RNA-seq; cultivar-by-infection interaction; preregistered analysis; cross-context evaluation

1. INTRODUCTION

Lychee (*Litchi chinensis* Sonn.) is a subtropical fruit crop of substantial economic importance in southern China, Southeast Asia, and a growing set of production regions worldwide [1]. Among its diseases, litchi downy blight, caused by the oomycete *Peronophythora litchii* (historically also treated as *Phytophthora litchii* [2]), is particularly damaging: the pathogen attacks young leaves, inflorescences, and fruit, produces water-soaked brown lesions, and drives severe postharvest decay [2,3]. Because chemical control is costly and resistance breeding depends on understanding how host genotypes differ in their response to infection, the molecular basis of cultivar-level variation in the response to *P. litchii* is a question of both agronomic and biological interest.

The resources needed to address this question computationally have accumulated rapidly. High-quality genome assemblies are available for lychee [4], and several groups have profiled the transcriptional response to *P. litchii* infection. Sun et al. compared a susceptible and a resistant cultivar across a pericarp infection time course and reported that the two genotypes deploy distinct early response programs [5]. Liu et al. described calcium-dependent protein kinase (LcCDPK) family members induced by *P. litchii* inoculation [6], and Li et al. showed that the pathogen RXLR effector PIAvh202 promotes virulence by destabilizing a host ethylene-biosynthesis enzyme [7]. A pathogen-side transcriptome catalogued the *P. litchii* pathogenicity arsenal [2]. In parallel, public repositories now hold RNA-seq series covering resistant and susceptible cultivars, leaf and fruit tissues, messenger and small RNA, and multiple time points. These series offer an opportunity, in principle, to distinguish genome-wide cultivar-dependent infection responses from responses common to all genotypes.

Realizing that opportunity is statistically demanding. The quantity of interest is not differential expression within a cultivar but the cultivar-by-infection interaction: the difference between infection responses across genotypes. Interaction effects are estimated with less precision than main effects, the deposited designs are small (typically three libraries per

cultivar–treatment cell), and a genome-wide search requires an explicit multiplicity family. Analyses that instead select a handful of strongly responding genes and then characterize them in the same cohort conflate selection with confirmation: large observed fold changes in a small experiment are a mixture of true signal and noise, and features chosen for extremity will regress toward null in any independent assessment. These concerns apply directly to the present pathosystem, in which most published candidate genes derive from single-cohort analyses without external evaluation.

Our own earlier work is part of that record. In an exploratory reanalysis of the GSE201243 leaf series [8] (Guiwei and Yurong1, mock versus inoculated, 24 h post-inoculation), we identified 17 significant infection-responsive genes in Guiwei and 117 in Yurong1, prioritized 18 candidates with defense-plausible annotations (among them jacalin-like lectin, pathogenesis-related and thaumatin/osmotin-like proteins, a heat-shock protein, lipid-transfer proteins, and calcium-binding proteins), and reported that a set of stress- and hormone-associated *cis*-regulatory elements was enriched in their promoters relative to a randomized background [9]. That analysis was framed as hypothesis-generating, and peer review of it converged on three substantive criticisms: the cultivar-by-treatment comparison was an effect-size contrast over pre-selected genes rather than a genome-wide interaction test with false-discovery-rate control; no independent dataset was used for validation; and promoter-element counting, even against a randomized background, is weak evidence of regulatory function. We regard all three criticisms as correct, and they motivated the confirmatory stage reported here.

This paper therefore presents the complete arc of the project in one place. We prospectively registered a discovery and external-evaluation protocol that locked dataset roles, models, thresholds, empty-result behavior, and the vocabulary used to describe evidence before any external outcome was opened. Because every cohort is public, this staged unlock is a computational safeguard enforced by the workflow rather than blinding in the experimental sense, and we document the registration chronology in the Methods. Within the locked discovery cohort we fitted a genome-wide negative-binomial interaction model, applied uniform mapping and gene-model quality control, and subjected the resulting candidates to a battery of internal robustness checks spanning independent quantification, an alternative statistical framework, filter sensitivity, and leave-one-library-out refits. We then evaluated the frozen candidates, a frozen pathway set, and a frozen signed expression signature in independent public cohorts whose roles (primary cross-context evaluation, generic transfer, orthogonal modality, or exploratory) were fixed in advance. Orthogonal evidence classes (sequence

95 annotation, promoter motifs, small-RNA coherence, and accession-aware literature) were kept
 96 separate from expression support, and a deterministic tier system integrated all layers while
 97 explicitly permitting an empty top tier. Finally, we audited the 18 exploratory candidates from the
 98 first stage against the genome-wide analysis. The result is a candidate resource for lychee
 99 downy blight whose every claim carries its evidence class, together with a case study in how
 100 much, and how little, of a plausible exploratory analysis survives formal statistical scrutiny.

101 2. RESULTS

102 2.1 A two-stage design with locked dataset roles

103 The study comprises an exploratory stage completed first and a confirmatory stage whose
 104 protocol was registered before its analyses began (Figure 1A; Table 1). Five public cohorts
 105 enter the confirmatory stage in fixed, non-interchangeable roles. PRJNA830488 (GSE201243),
 106 the leaf series for Guiwei and Yurong1 at 24 h, is the sole discovery cohort; it is also the dataset
 107 analyzed in the exploratory stage. PRJNA450886, a pericarp time course contrasting the
 108 susceptible cultivar Guiwei with the resistant cultivar Heiye at 6, 24, and 48 h [5], serves as the
 109 primary external evaluation. Because it differs from the discovery cohort in tissue, resistant
 110 comparator, and time structure, we describe positive outcomes in it as **cross-context support**
 111 rather than replication. PRJNA922966 (Feizixiao leaf and fruit, mock versus infected) tests
 112 generic infection-response transfer but cannot address the cultivar interaction; PRJNA922965
 113 provides small-RNA libraries as an orthogonal modality; and PRJNA1090613 (Guiwei versus
 114 SFZ leaf) is exploratory only, because its time and resistance metadata could not be resolved.
 115 Under the registered firewall, no external outcome could alter the frozen discovery membership,
 116 signature weights, or thresholds, and the protocol explicitly allowed the top evidence tier to be
 117 empty.

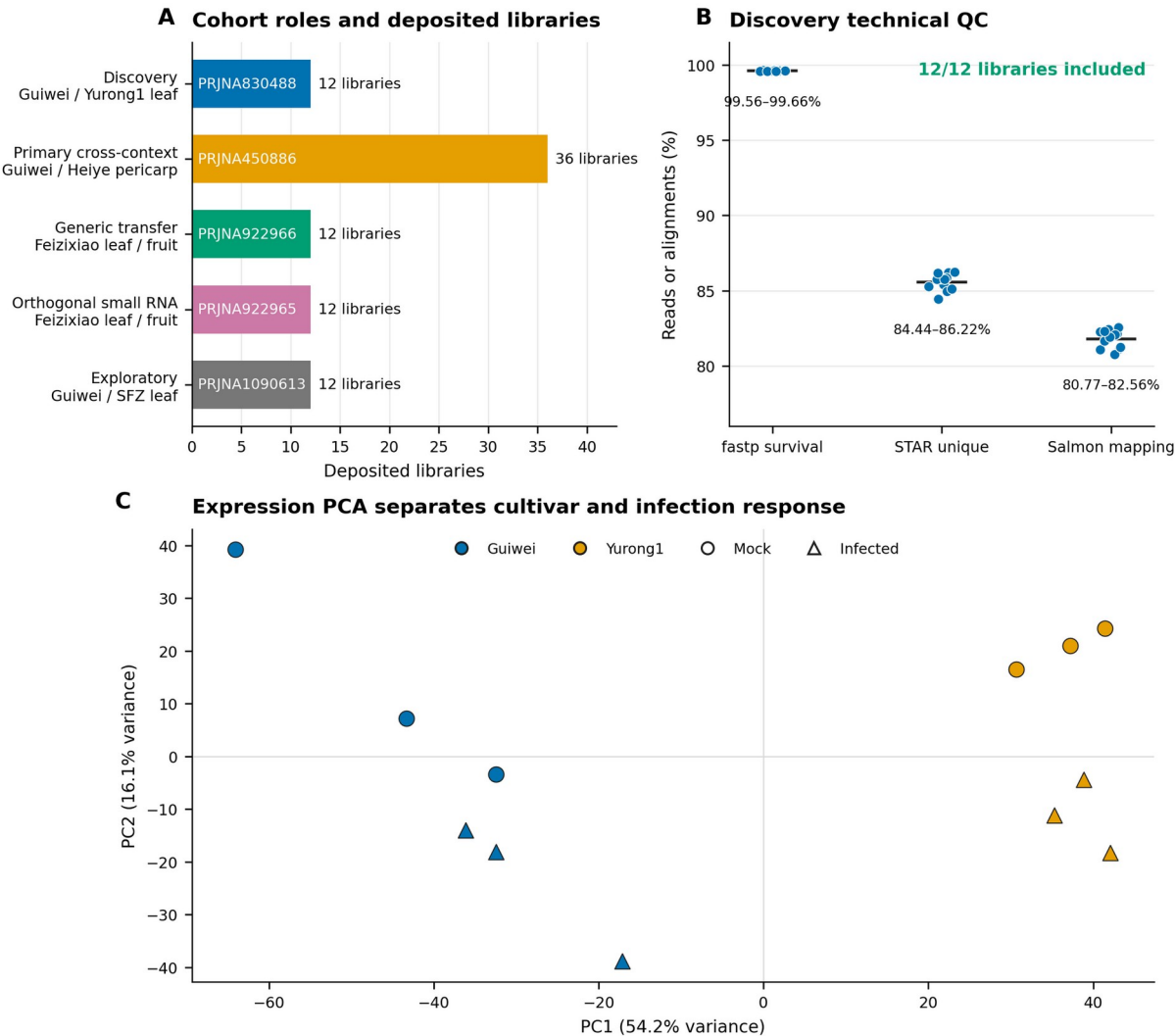


Figure 1. Study cohorts, discovery quality control, and expression structure. (A) The five public cohorts with locked roles and deposited library counts. (B) Per-library technical quality in the discovery cohort: fastp read survival, STAR unique-alignment rate, and Salmon mapping rate; all 12 libraries passed the fixed inclusion gates. (C) Principal-component analysis of the discovery libraries separates cultivar and infection status.

****Table 1. Locked dataset roles and eligibility.****

Accession	Design	Locked role	Evidence class		
PRJNA830488					
(GSE201243)	Guiwei/Yurong1 leaf; mock/infected; 24 h; 3 libraries per cell	Discovery			
	Conditional inferential				
PRJNA450886	Guiwei/Heiye pericarp; mock/infected; 6, 24, 48 h; 3 libraries per cell	Primary external evaluation	Cross-context		
PRJNA922966	(GSE222651) Feizixiao leaf/fruit; mock/infected; 24 h; 3 libraries per cell	Generic infection transfer	Cross-context		
PRJNA922965	(GSE222650) Feizixiao leaf/fruit small RNA; mock/infected; 24 h; 3 libraries per cell	Orthogonal modality	Orthogonal		
PRJNA1090613	(GSE262200) Guiwei/SFZ leaf; mock/infected; 3 libraries per cell	Exploratory only	Exploratory		

2.2 Stage 1: exploratory candidate discovery in GSE201243

The exploratory stage analyzed the 12 discovery libraries with per-cultivar differential-expression contrasts (infected versus mock) and identified 17 significant genes in Guiwei (15 up-regulated, 2 down-regulated) and 117 in Yurong1 (39 up-regulated, 78 down-regulated) at Benjamini–Hochberg adjusted $P^* < 0.05$, suggesting a broader transcriptional response in Yurong1. From these lists, 18 genes were prioritized by combining statistical significance in at least one cultivar, large infection-associated fold changes, and annotations plausibly related to defense: among them a jacalin-like lectin (LITCHI028104), a pathogenesis-related protein (LITCHI011394), osmotin/thaumatin-like proteins (LITCHI009301, LITCHI002793), an HSP70-family chaperone (LITCHI012388), lipid-transfer proteins (LITCHI019183, LITCHI018043), an EF-hand calcium-binding protein (LITCHI019519), and a phytosulfokine precursor (LITCHI007102). A selected-gene contrast of infection responses between cultivars nominated LITCHI019519, LITCHI001510, LITCHI028401, and LITCHI017676 as Guiwei-skewed and LITCHI028104 and LITCHI019183 as Yurong1-skewed candidates.

The exploratory stage also examined 2-kb upstream promoter regions of the 18 candidates. Against 100,000 randomized GC-matched promoter sets, four *cis*-element classes appeared enriched: anaerobic-response elements (ARE; 46 observed versus 24.6 expected, $q^* = 0.0003$), methyl-jasmonate-responsive motifs (54 versus 38.4, $q^* = 0.018$), TCA elements (9 versus 0.19, $q^* = 0.00003$), and TC-rich repeats (10 versus 0.15, $q^* = 0.00003$). ABRE and MYB-binding-site motifs, in contrast, were frequent but not enriched. All of these results were obtained by selecting and characterizing genes within a single cohort; they carry no genome-wide multiplicity control and no external evaluation, and we treat them below strictly as hypotheses submitted to the confirmatory stage. Full exploratory-stage figures, tables, and workflow scripts are available with the original report [9].

2.3 Discovery-cohort quality under the registered protocol

The confirmatory pipeline combined a joint host and pathogen reference, fastp trimming, STAR two-pass alignment, featureCounts gene counting, and independent Salmon quantification. Under this pipeline, all 12 discovery libraries passed the fixed technical gates of at least 10 million surviving read pairs and at least 40% uniquely aligned reads (Figure 1B). Read survival after trimming ranged from 99.56% to 99.66%, unique-alignment rates from 84.44% to 86.22%, and Salmon mapping rates from 80.77% to 82.56%. Principal-component analysis of the transformed counts separated libraries by cultivar on the first component (54.2% of variance) and by infection status on the second (16.1%), with no library isolated from its replicate group (Figure 1C). Because source-tree, pooling, and extraction independence are not

166 reported for the deposited libraries, all inferential statements below are conditional on
 167 deposited-library independence, a limitation we return to in the Discussion.

168 **2.4 Genome-wide cultivar-by-infection discovery**

169 The primary discovery model fitted all expressed genes with the design $\sim$ cultivar +
 170 treatment + cultivar:treatment, with Guiwei and mock as reference levels, so that the interaction
 171 coefficient estimates how much the infected-minus-mock response in Yurong1 exceeds that in
 172 Guiwei. Of 19,445 expressed genes, 262 met the registered discovery threshold of genome-
 173 wide Benjamini–Hochberg $q^* < 0.05$ with an absolute interaction $\log_2$ fold change of at least
 174 $\log_2(1.5)$ (Figure 2A); 277 genes met the significance criterion alone and 5,370 met the effect
 175 criterion alone. All interaction estimates reported and plotted here are unshrunk maximum-
 176 likelihood values; apeglm shrinkage was used only to define the frozen signature weights.
 177 Uniform quality control then removed 31 candidates for exon-level mappability failure and 25 for
 178 gene-model ambiguity, freezing 206 genes as statistical candidates (Figure 2C). Because these
 179 quality filters were applied after significance testing, we verified that the ordering is not load-
 180 bearing: restricting the testing universe to the 13,602 genes that pass uniform quality control
 181 and recomputing the Benjamini–Hochberg adjustment over that universe rediscovers all 206
 182 frozen candidates and adds 12 further genes, so the frozen list is a conservative subset of the
 183 QC-first analysis. A post hoc composite-null sensitivity analysis asked the stricter question
 184 whether $|\beta|$ exceeded $\log_2(1.5)$, rather than testing $\beta = 0$ and filtering the estimate afterward.
 185 Eight of the 262 statistical candidates passed the DESeq2 composite-null test at $q^* < 0.05$; the
 186 corresponding counts were 7 of 206 QC-retained candidates, 7 of 19 edgeR-gated candidates,
 187 and 7 of 16 fully robust candidates. An apeglm false-sign-or-small analysis was less severe,
 188 with $s^* < 0.05$ for 149, 113, 18, and 15 genes in those four nested sets, respectively. These are
 189 post hoc sensitivity results, not a redefinition of the registered discovery set. Positive interaction
 190 effects (stronger infection response in Yurong1) and negative effects (stronger response in
 191 Guiwei) were both well represented. At the pathway level, six Plant Reactome pathways
 192 transported through one-to-one *Oryza** orthology met the frozen discovery threshold: activation
 193 of the pre-replication complex, assembly of the pre-replication complex, DNA replication
 194 initiation, maturation, and circadian rhythm, all with positive enrichment, and jasmonic acid
 195 signaling with negative enrichment. Together with the genes, these six pathways, the apeglm-
 196 shrunken signed signature weights over the 206 genes, and the conditional transcript-usage
 197 results (Section 2.8) were frozen before any external data were opened.

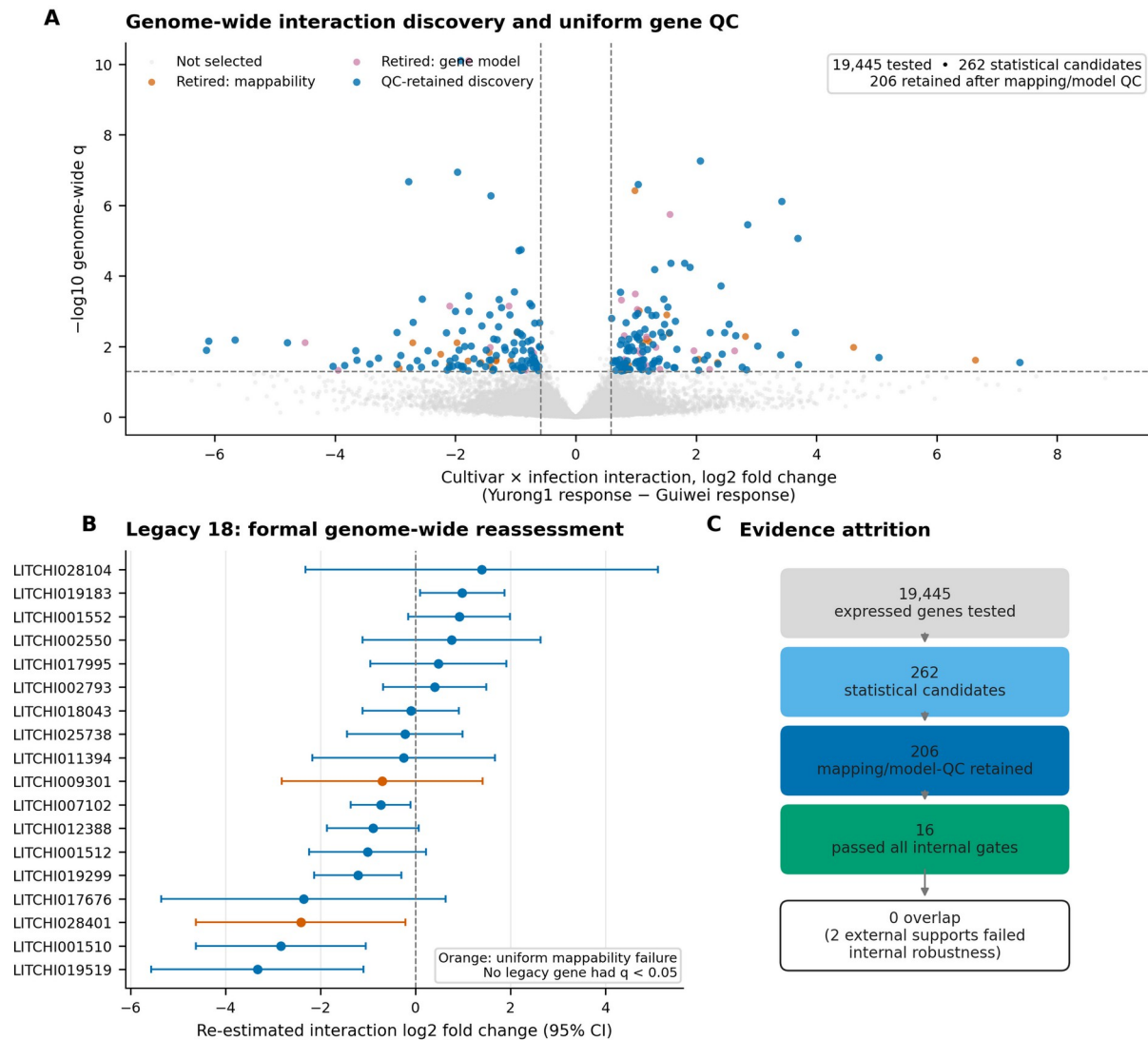


Figure 2. Genome-wide discovery and formal reassessment of the exploratory candidates. (A) Cultivar-by-infection interaction effects across 19,445 expressed genes; 262 genes met the statistical threshold and 206 were retained after uniform mapping and gene-model quality control. (B) Genome-wide re-estimation of the 18 exploratory (legacy) candidates with 95% confidence intervals; orange marks uniform-mappability failures. No legacy candidate met the genome-wide interaction criterion. (C) Evidence attrition from tested genes to internally robust candidates.

2.5 None of the 18 exploratory candidates met the genome-wide interaction criterion

Because the exploratory candidates were selected before any interaction P^* -values existed, the registered protocol included a formal audit of all 18 against the genome-wide analysis (Figure 2B; Table 2). None met the genome-wide interaction threshold. We then performed the estimand-aligned follow-up requested in review: separate confirmatory DESeq2 infected-versus-mock fits within each cultivar, with Benjamini–Hochberg adjustment over the same 19,445-gene universe in each contrast (Supplementary S14). Thirteen of the 18 legacy genes were significant in Guiwei and 16 of 18 in Yurong1 at $q^* < 0.05$; all 13 Guiwei genes and 15 of the 16 Yurong1 genes also exceeded $|\log_2FC| \geq \log_2(1.5)$. Thus most of the original

within-cultivar response claims survive the confirmatory pipeline even though none supports a genome-wide cultivar-by-infection interaction. The strongest interaction-audit candidate, LITCHI001510, reached $q^* = 0.080$; the EF-hand calcium-binding gene LITCHI019519, whose selected-gene contrast of -3.4 had appeared most dramatic in the exploratory stage, reached only $q^* = 0.106$ despite a re-estimated interaction effect of -3.33 ; and the median genome-wide interaction q^* across the 18 candidates was 0.47 . Two candidates, the osmotin/thaumatin-like gene LITCHI009301 and LITCHI028401, additionally failed uniform exon-mappability control, meaning their interaction estimates cannot be attributed unambiguously to a single locus. The audit therefore does not show that these genes are biologically inert. It shows that within-cultivar responsiveness and cultivar-dependent response are distinct claims, and only the latter fails here.

Table 2. Formal genome-wide audit of the 18 exploratory-stage candidates. Interaction effects are re-estimated genome-wide values (Yurong1 response minus Guiwei response, log2 scale). No candidate met the registered discovery threshold (genome-wide $q^* < 0.05$ and $|\log_2FC| \geq \log_2(1.5)$).

Gene	Exploratory annotation	Interaction log2FC (SE)	Genome-wide q^*	Uniform mappability
LITCHI001510	Poorly characterized responsive gene	-2.83 (0.91)	0.080	Pass
LITCHI019519	EF-hand calcium-binding protein	-3.33 (1.14)	0.106	Pass
LITCHI019299	S-adenosylmethionine-related protein	-1.21 (0.47)	0.175	Pass
LITCHI007102	Phytosulfokine precursor	-0.74 (0.32)	0.238	Pass
LITCHI019183	Plant lipid-transfer protein	$+0.99$ (0.45)	0.273	Pass
LITCHI028401	Selected interaction candidate	-2.41 (1.13)	0.281	Fail
LITCHI012388	Heat-shock protein 70	-0.90 (0.49)	0.381	Pass
LITCHI001552	Cyclase-like stress-related protein	$+0.92$ (0.55)	0.430	Pass
LITCHI001512	Poorly characterized responsive gene	-1.01 (0.63)	0.457	Pass
LITCHI017676	S-adenosylmethionine-related protein	-2.36 (1.53)	0.478	Pass
LITCHI002550	Cytochrome P450	$+0.76$ (0.96)	0.762	Pass
LITCHI028104	Jacalin-like lectin	$+1.39$ (1.89)	0.785	Pass
LITCHI002793	Osmotin/thaumatin-like protein	$+0.41$ (0.55)	0.786	Pass
LITCHI017995	Metalloendoproteinase	$+0.48$ (0.73)	0.812	Pass
LITCHI009301	Osmotin/thaumatin-like protein	-0.70 (1.08)	0.816	Fail
LITCHI025738	Cyclase-like stress-related protein	-0.22 (0.62)	0.908	Pass
LITCHI011394	Pathogenesis-related protein	-0.25 (0.98)	0.940	Pass
LITCHI018043	Plant lipid-transfer protein	-0.10 (0.52)	0.954	Pass

2.6 Sixteen genes and one pathway passed all internal robustness gates

Each of the 206 frozen genes was required to retain its interaction direction and significance under four predeclared perturbations of the analysis: independent Salmon-based quantification (196 of 206 passed), edgeR quasi-likelihood testing in place of the primary framework (19 passed), a counts-per-million expression filter (205 passed), and all twelve leave-one-library-out refits (125 passed). Eighteen genes passed all four gates jointly, and 16 remained after an additional observed mapping-sensitivity gate that excluded candidates whose effect depended on ambiguously mapped reads (Figure 3A). The alternative-framework gate was by far the most stringent, and the frozen genome-wide edgeR results explain why. The two frameworks agree almost perfectly on estimation: interaction effects correlate at Pearson $r^* = 0.995$ across all 19,445 genes and $r^* = 0.999$ across the 206 candidates, sign agreement among the candidates is 100%, and every candidate has nominal edgeR $P^* < 0.05$. The disagreement is confined to significance calibration. The registered gate required genome-wide edgeR $q^* < 0.10$ with direction agreement, and edgeR's quasi-likelihood test with robust dispersion estimation admits only 24 genes genome-wide at that threshold (10 at $q^* < 0.05$), against 262 for the primary Wald analysis at $q^* < 0.05$. With three libraries per cell, the two dispersion-estimation strategies imply very different genome-wide significance sets even while agreeing on every point estimate. We therefore present the results as a hierarchy: 262 statistical candidates, 206 retained after quality control, 19 significant under both frameworks, and 16 passing the complete registered procedure; genes surviving both frameworks constitute the discovery set's stable core, and the wider 206-gene list should be read as one framework's candidates rather than as settled discoveries. Of the six frozen pathways, only circadian rhythm passed its four conjunctive gates (competitive and rotation-based testing, leading-edge deletion, and an expression- and length-matched null); the other five all failed the rotation-based self-contained test, and the maturation set additionally failed leading-edge deletion, indicating enrichment that was not stable across testing frameworks.

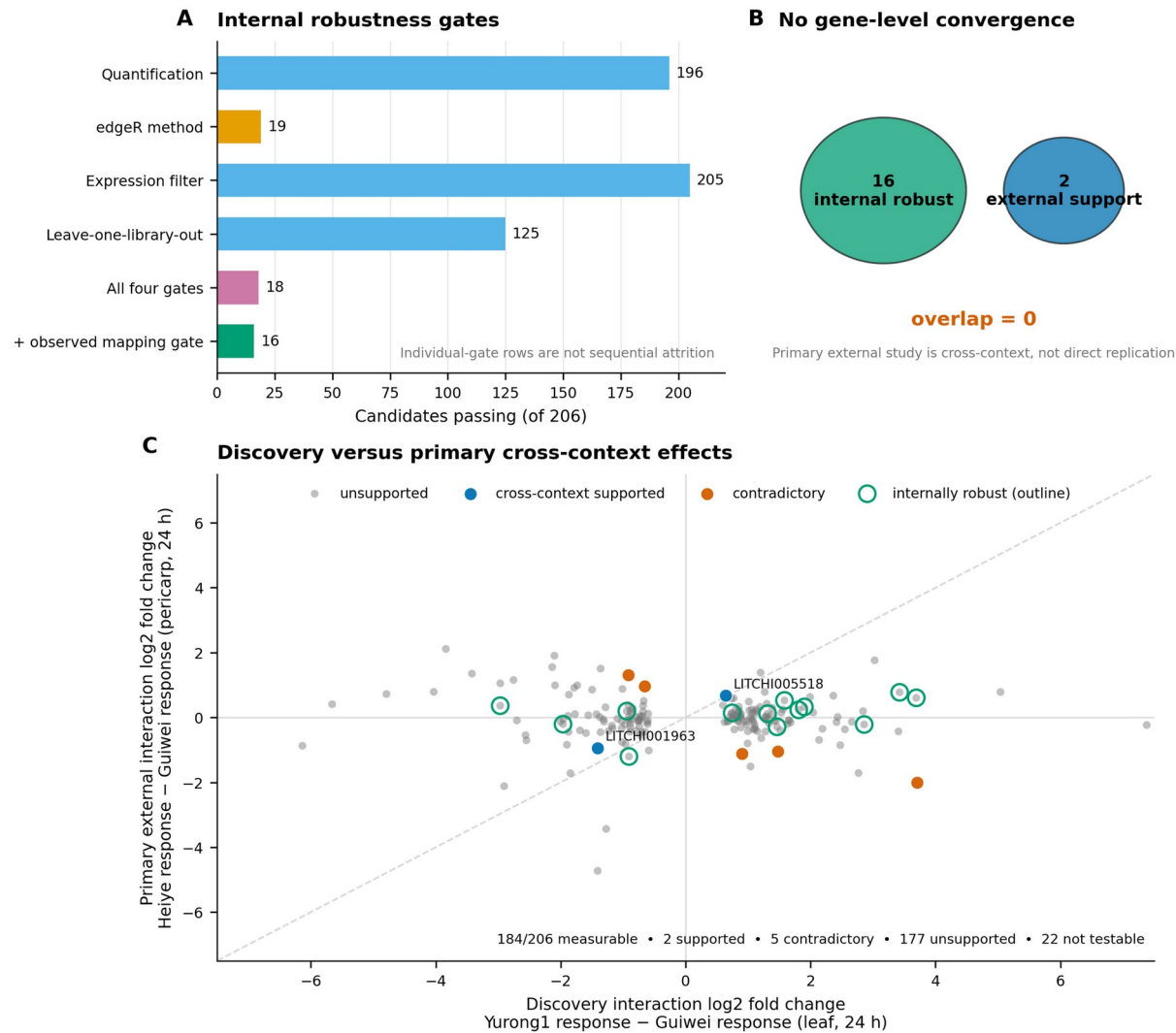


Figure 3. Internal robustness and the primary external evaluation. (A) Candidates passing each predeclared robustness gate, the conjunction of the four analytic gates, and the additional observed-mapping gate. (B) The internally robust set (16 genes) and the externally supported set (2 genes) do not overlap. (C) Discovery interaction effects against the primary cross-context interaction effects (Heiye response minus Guiwei response, pericarp, 24 h); supported, contradictory, and internally robust genes are highlighted. The diagonal is a visual reference only; equal effects across contexts are not expected.

2.7 The locked external evaluation supported two genes and contradicted five

The primary external estimand, fixed before unlock, was the resistant-minus-susceptible cultivar-by-infection interaction at 24 h in the PRJNA450886 pericarp time course, tested for each frozen candidate with Benjamini–Hochberg adjustment across the full candidate-by-contrast family, an effect threshold of $\log_2(1.5)$, confidence intervals excluding zero, and required agreement with the discovery direction. Of the 206 frozen genes, 184 were measurable in the external cohort. Two met every criterion and are cross-context supported: LITCHI005518, annotated as an alpha-xylosidase involved in xyloglucan remodeling of the cell wall, and LITCHI001963, an uncharacterized gene with a one-to-one *Oryza* ortholog. Five genes,

including a beta-glucosidase and a Fe(II)/2-oxoglutarate-dependent dioxygenase, passed the magnitude and significance criteria with the *opposite* direction and are recorded as contradictory; 177 were unsupported at the prespecified thresholds and 22 not testable (Figure 3C; Table 3). At the level of continuous estimates, discovery and external interaction effects across the 184 measurable candidates were uncorrelated (Pearson $r^* = -0.05$, 95% CI -0.19 to 0.10 ; Spearman $\rho = -0.04$), with sign concordance of 51.6% (95 of 184), indistinguishable from chance.

The two supported genes are not among the 16 internally robust genes, and none of the 16 internally robust genes achieved external support (Figure 3B). This non-overlap is, by itself, nearly uninformative: for sets of 16 and 2 genes drawn from 206 candidates, the expected overlap under independence is 0.16 genes and the probability of zero overlap is 0.85, so the disjoint sets neither demonstrate nor refute an association between internal stability and cross-context transport. The absence of continuous concordance reported above is the more informative observation. We emphasize that the external cohort differs in tissue (pericarp versus leaf), resistant comparator (Heiye versus Yurong1), and design (time course versus single time point), so absence of support is expected wherever a response is tissue- or genotype-specific, and support that is observed indicates transport across contexts rather than replication of the original experiment. Because Yurong1 and Heiye have not been placed on a common resistance scale relative to Guiwei, we interpret the primary external test as cross-cultivar response transport rather than validation of a resistance phenotype.

Table 3. Cross-context supported and contradictory genes in the primary external evaluation (PRJNA450886, pericarp, 24 h). Discovery effects are Yurong1-minus-Guiwei interaction log2 fold changes in leaf; external effects are Heiye-minus-Guiwei interaction log2 fold changes in pericarp with 95% confidence intervals. All q^* -values are Benjamini–Hochberg adjusted (discovery: genome-wide; external: across the frozen candidate-by-contrast family).

Gene	Annotation	Discovery log2FC	Discovery q^*	External log2FC (95% CI)	External q^*	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterized; one-to-one <i>Oryza</i> * ortholog	-1.41	0.024	-0.94 (-1.43 to -0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	-0.92	0.040	+1.31 (+0.78 to +1.85)	0.0002	Contradictory
LITCHI029534	Thiamine thiazole synthase 2, chloroplastic	+1.48	0.002	-1.04 (-1.61 to -0.48)	0.010	Contradictory
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	-0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Contradictory
LITCHI008313	Beta-glucosidase 44					

320 +0.90 | 0.045 | -1.12 (-1.81 to -0.44) | 0.034 | Contradictory | | LITCHI001700 | Fe(II)/2-
 321 oxoglutarate-dependent dioxygenase 4 | +3.70 | 0.033 | -2.00 (-3.24 to -0.77) | 0.035 |
 322 Contradictory |

323 **2.8 Pathway and signature transport was weak and strongly context-dependent**

324 None of the six frozen pathways, including the internally robust circadian-rhythm pathway,
 325 passed the external pathway gates in PRJNA450886 (Figure 4A). The frozen signed signature
 326 over the 206 discovery genes behaved in a strongly context-dependent way (Figure 4B). At the
 327 primary 24-h contrast the signature score was indistinguishable from zero (estimate 0.109, 95%
 328 CI -0.079 to 0.297, $q^* = 0.349$), and at 48 h it was significantly *negative* (-0.240, 95% CI
 329 -0.428 to -0.052, $q^* = 0.043$), a sign reversal relative to discovery. In the generic-transfer
 330 study the signature tracked infection positively in leaf (0.107, $q^* = 0.0015$) but negatively in fruit
 331 (-0.181, $q^* < 0.001$), with a significant leaf-minus-fruit interaction (0.289, $q^* < 0.001$). The
 332 signature therefore carries reproducible infection-associated information, but its sign and
 333 magnitude depend on tissue and time in ways that preclude its use as a context-free resistance
 334 marker. The exploratory cohort's estimate is reported separately in Figure S3 for completeness
 335 and is not confirmatory.

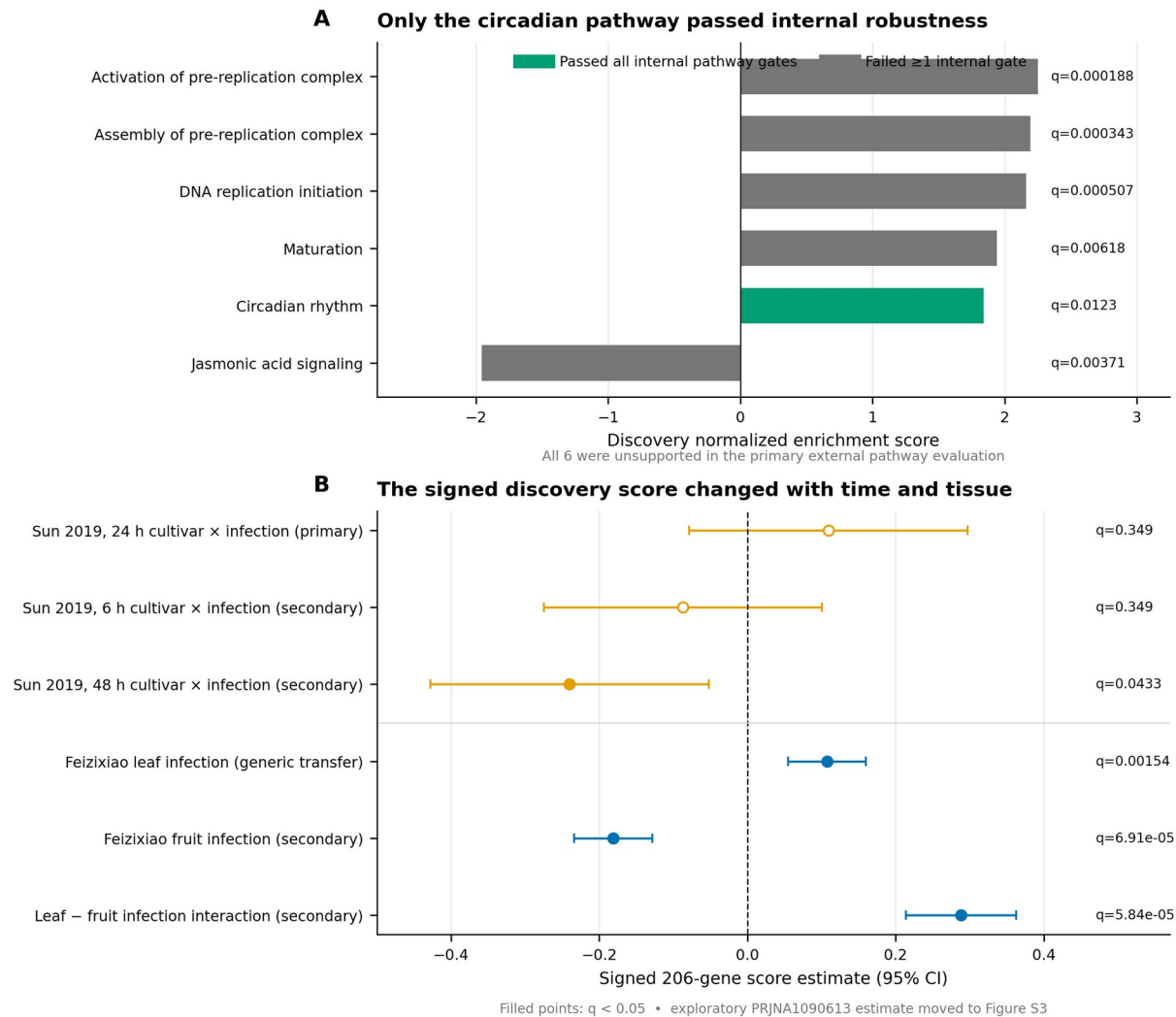
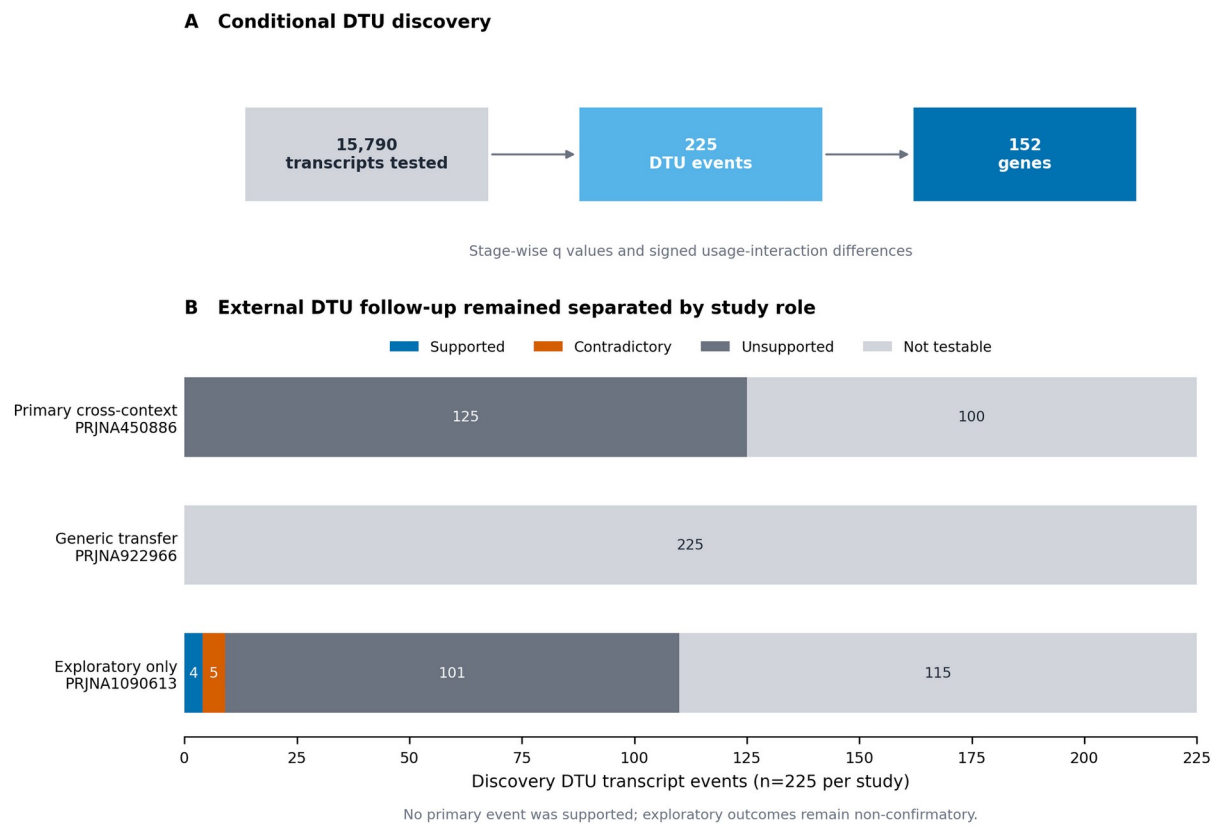


Figure 4. Frozen pathway and signature evaluation. (A) Discovery normalized enrichment scores for the six frozen pathways; only circadian rhythm passed all internal gates, and none was supported in the primary external evaluation. (B) The frozen signed 206-gene signature across the six non-exploratory external contrasts; filled points denote $q < 0.05$. The quarantined PRJNA1090613 estimate is in Figure S3.

2.9 Conditional transcript-usage discovery remained internal

Differential transcript usage was analyzed under a conditional gate registered in advance: stage-wise testing had to yield interpretable gene- and transcript-level error control, and no post hoc substitute was permitted if the gate failed. The gate passed, and 225 transcript-usage events across 152 genes were frozen from 15,790 tested transcripts (Figure 5A). External follow-up preserved the role separation of the cohorts (Figure 5B): in the primary cross-context study, 125 events were measurable but unsupported at the prespecified thresholds and 100 were not testable; in the generic-transfer study, annotation incompatibilities left all 225 untestable; and in the exploratory cohort, 4 events were supported and 5 contradictory; we report these outcomes descriptively and do not count them as primary external support,

351 because that cohort's metadata could not be standardized. Transcript-level rewiring between
352 cultivars therefore remains an internally supported, externally unconfirmed layer of the resource.

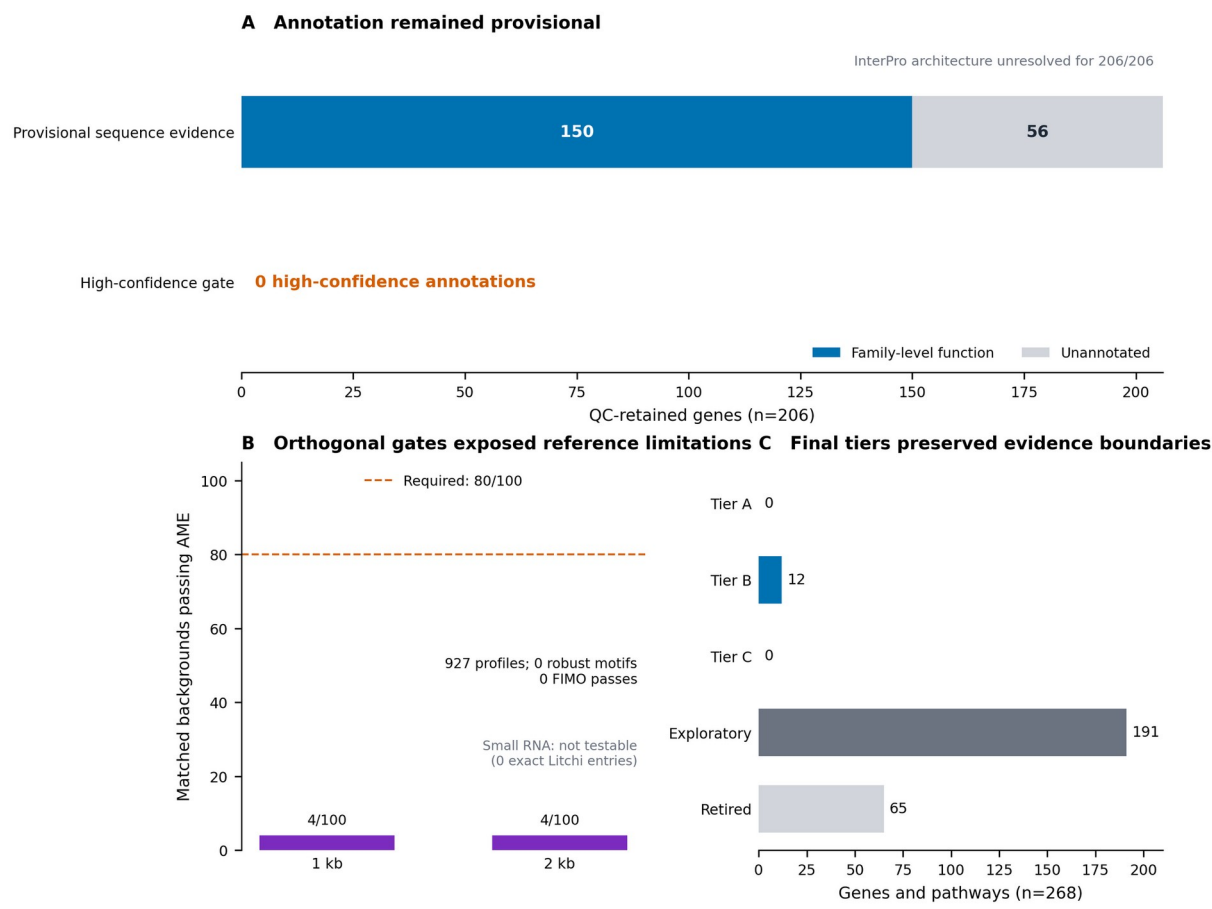


353 Figure 5. Conditional transcript usage. (A) Discovery gate and event counts. (B) External follow-up of the 225
354 discovery events, separated by locked study role.
355

356 **2.10 Orthogonal evidence: informative nulls and reference-ecosystem gaps**

357 The registered orthogonal evidence classes returned uniformly negative or non-testable
358 results, and we report them as such rather than converting them into support. Computational
359 annotation assigned family-level functional labels to 150 of the 206 candidates and left 56
360 unannotated, but no candidate met the registered high-confidence rule requiring two
361 independent evidence classes with at least 70% supported coverage and no architecture
362 conflict, largely because precomputed InterPro domain architectures were unavailable for all
363 206 proteins in this non-model species (Figure 6A). Registered promoter-motif discovery tested
364 927 JASPAR plant position-weight matrices in strand-aware 1-kb and 2-kb windows against 100
365 expression- and GC-matched genomic background sets. No motif passed in more than 4 of 100
366 backgrounds against a required 80, and none passed FIMO sensitivity (Figure 6B). Small-RNA
367 coherence was not testable because the frozen PmiREN reference contained no exact *Litchi*
368 entries.

369 A separate post hoc controlled comparison held the published 18-promoter totals, six
370 explicitly reported exact-element strings plus one standard PlantCARE TCA-element variant,
371 scoring rule, and multiplicity correction fixed while changing only the background
372 (Supplementary S15). The 100,000-set randomized-GC rerun closely reproduced the
373 exploratory result. Against 100 expression- and GC-matched genomic promoter sets, ARE lost
374 significance, ABRE gained it, and MeJA-responsive, TCA, and TC-rich classes remained
375 significant: four of seven classes passed under either strategy, but the identity of one class
376 changed. Background construction therefore affected the result but did not by itself explain the
377 exploratory/registered discrepancy. This comparison is provenance-limited: the archived
378 exploratory package omitted the observed-motif input TSV, and exact recounting from the
379 current canonical 18 promoters reproduced the published total only for ARE. Accordingly, it
380 cannot rescue a functional regulatory claim or establish which foreground representation
381 generated the other published totals. Published studies reusing any analyzed accession,
382 including the primary external study itself [5], remain registered as prior interpretation rather
383 than independent support.



385 Figure 6. Orthogonal evidence and final tiers. (A) Annotation status of the 206 QC-retained genes. (B) Registered
 386 promoter-motif robustness and the small-RNA reference gate. (C) Deterministic final tiers over all 268 frozen entities.

387 **2.11 Deterministic integration: an empty top tier, twelve Tier B genes, and visible retirements**

388 The registered tier rules integrate all evidence layers without discretion. Tier A required
 389 discovery, complete internal robustness, primary cross-context support, clean mapping and
 390 annotation, and at least one attributable orthogonal class; Tier B required discovery and full
 391 internal robustness with partial or non-testable external evidence; Tier C was reserved for a
 392 robust, externally supported pathway. Applied to the 268 frozen entities (262 statistical gene
 393 candidates and 6 pathways), the rules assigned no entity to Tier A or Tier C, 12 genes to Tier B,
 394 191 entities to exploratory status, and 65 to retirement (Figure 6C). The retirements comprise
 395 the 56 statistical candidates removed by uniform mapping or gene-model control, the 5 genes
 396 contradicted for cross-context directional transport, and 4 candidates with candidate-level
 397 observed-mapping or annotation failures. These categories differ in kind, and the released
 398 tables preserve the distinction: mapping and annotation failures undermine the discovery itself,
 399 whereas an opposite direction in a different tissue and cultivar contrast may reflect genuine
 400 context specificity rather than an invalid discovery in the original context. Four internally robust
 401 genes, including a probable RPP8-like disease-resistance protein, were measurable but
 402 unsupported at the prespecified thresholds in the primary external evaluation and were
 403 therefore held at exploratory status rather than promoted. The two cross-context-supported
 404 genes likewise remain exploratory because they failed internal robustness. All contradictory and
 405 failed outcomes are released alongside the positive ones. We also note that Tier A was difficult
 406 to reach by construction in the current reference ecosystem: with the small-RNA class
 407 structurally untestable, no motif able to pass the family-wise rule, accession-linked literature
 408 excluded as independent support, and the two-class annotation rule unattainable without
 409 precomputed domain architectures, the orthogonal-class requirement could not realistically be
 410 satisfied by any gene. The empty top tier therefore reflects this structural constraint as much as
 411 the expression evidence itself.

412 The 12 Tier B genes are listed in Table 4. Nine were measurable in the primary external
 413 cohort, and all nine were direction-consistent with discovery, although none met the full support
 414 criteria. Figure S1 exposes the normalized count of every individual discovery library for these
 415 12 genes, the 2 cross-context-supported genes, and legacy highlights LITCHI001510 and
 416 LITCHI019519; every cultivar-treatment cell contains all three deposited replicates. Their
 417 annotations cluster around carbohydrate and cell-wall metabolism (a
 418 phosphomannomutase/phosphoglucomutase, a chloroplastic 1,4-alpha-glucan-branching

enzyme, an ERD6-like sugar transporter, a serine carboxypeptidase), membrane transport (a WAT1-related protein), lipid signaling (a DIR1-like lipid-transfer protein, the **Arabidopsis** ortholog of which is required for systemic acquired resistance), programmed-cell-death regulation (a BAG-family chaperone regulator), specialized metabolism (cytochrome P450 81Q32), RNA metabolism (ribonuclease J, exonuclease 1), chromatin-associated regulation (a homeobox-DDT protein), and one entirely unannotated gene.

****Table 4. The twelve Tier B genes: internally robust genome-wide statistical candidates with direction-consistent but sub-threshold, or non-measurable, primary external evidence.**** Discovery effects are interaction log2 fold changes (Yurong1 response minus Guiwei response); positive values indicate a stronger infection response in Yurong1. External columns give the primary 24-h pericarp estimate with 95% confidence interval and family-adjusted **q**; NM denotes genes not measurable in the external cohort.

Gene	Family-level annotation	Discovery log2FC	Genome-wide <i>*q*</i>	External log2FC (95% CI)	External <i>*q*</i>
LITCHI014613	Sugar transporter ERD6-like 16	+3.69	8.7×10^{-6}	+0.60 (−0.04 to +1.25)	0.27
LITCHI015982	WAT1-related protein	+3.43	7.8×10^{-7}	+0.77 (+0.07 to +1.48)	0.18
LITCHI005805	Phosphomannomutase/phosphoglucomutase	+2.07	5.5×10^{-8}	NM	—
LITCHI028926	Exonuclease 1	+1.90	5.7×10^{-5}	+0.32 (−0.63 to +1.28)	0.83
LITCHI016077	Serine carboxypeptidase 24	+1.81	4.4×10^{-5}	+0.23 (−1.05 to +1.51)	0.89
LITCHI013915	1,4-alpha-glucan-branching enzyme 3, chloroplastic	+1.58	4.4×10^{-5}	+0.53 (+0.10 to +0.95)	0.13
LITCHI021546	Homeobox-DDT domain protein RLT3	+1.31	6.6×10^{-5}	+0.12 (−0.30 to +0.54)	0.83
LITCHI022605	Ribonuclease J	+0.74	2.9×10^{-4}	+0.14 (−0.34 to +0.62)	0.83
LITCHI028717	BAG-family molecular chaperone regulator 4	−0.91	1.8×10^{-5}	−1.20 (−2.52 to +0.12)	0.29
LITCHI008721	Unannotated protein	−1.91	7.9×10^{-11}	NM	—
LITCHI015740	Putative lipid-transfer protein DIR1	−1.96	1.2×10^{-7}	−0.21 (−1.10 to +0.69)	0.88
LITCHI010877	Cytochrome P450 81Q32	−2.77	2.1×10^{-7}	NM	—

2.12 Parametric simulations quantify the interaction effects detectable with three libraries per cell

The post hoc power analysis simulated negative-binomial counts from the fitted gene-wise means and dispersions while preserving the four-cell discovery design and the 24-h four-cell external design (Figure S2; Supplementary S18). At each of 12 absolute interaction effects from 0 to 4 log2 units, 100 simulations were run sequentially. Genome-wide discovery used 19,445 tests with 262 injected targets per simulation, matching the observed statistical-candidate prevalence; the external analysis adjusted within the 177 frozen genes that passed the

simulation's count and dispersion validity filters, with 2 injected targets matching the number of directionally supported genes. Seven additional genes were measurable in the observed 184-gene external contrast but were not fit-eligible for simulation, so the simulated multiplicity family is reported explicitly rather than equated with the observed family. The overall interpolated effect required for 80% detection was 2.44 log₂FC (5.4-fold) in discovery and 2.11 log₂FC (4.3-fold) in the external candidate-family analysis. At a true |interaction log₂FC| of 1.5, overall detection probabilities were 62.7% and 74.0%, respectively. These values are conditional on fitted dispersions treated as known, deposited-library independence, and the simulated non-null prevalence; they quantify the low-power regime but cannot distinguish context dependence from absence of effect for any particular unsupported gene.

3. DISCUSSION

The central finding of this unified analysis is methodological and biological at once: in the lychee–**P. litchii** pathosystem, the genes that look most impressive in a single-cohort exploratory analysis and the genes that survive formal genome-wide inference are almost entirely different sets. The 18 exploratory candidates were chosen for large fold changes and defense-plausible annotations, yet none reached genome-wide significance for the cultivar-by-infection interaction, and two could not even be attributed unambiguously to a single locus. Conversely, the 16 genes that passed every internal robustness gate, and the 12 that reached Tier B, are dominated not by canonical pathogenesis-related families but by carbohydrate and cell-wall metabolism, transport, lipid signaling, and RNA metabolism, including one gene with no annotation at all. Neither observation would have been visible without running both stages, and together they quantify a selection effect that is likely to operate in any small-replicate transcriptomic study of this kind. One qualification applies throughout: the audit changed not only the statistical machinery but, for many of the 18 genes, the scientific question, from within-cultivar responsiveness to cultivar-dependent response, so the comparison measures survival under a stricter and partly different criterion rather than direct refutation of the original claims.

Why did the exploratory candidates fail the interaction audit? Three mechanisms, all anticipated by the reviewers of the exploratory report, are sufficient to explain the outcome. First, selection on observed fold change in a three-replicate contrast preferentially captures genes with high sampling variance, and their apparent effects regress when the full uncertainty is modeled: LITCHI019519's selected-gene contrast of –3.4 looked decisive, but its genome-wide interaction test carried a standard error of 1.14 and a *q*-value of 0.106. Second, per-cultivar differential expression followed by informal comparison is not an interaction test; genes

can differ dramatically in within-cultivar significance while their responses do not differ significantly between cultivars. Third, two candidates failed uniform-mappability control, a failure mode invisible to any analysis that trusts a single quantification path. None of this implies the exploratory candidates are biologically irrelevant; several retain large direction-consistent estimates and remain reasonable targets for quantitative PCR. Their evidentiary status, however, is now stated correctly: unconfirmed hypotheses, not validated markers.

The annotations of the genes that survive scrutiny suggest a recognizable, although still hypothetical, theme. Annotations related to cell-wall and sugar metabolism appear at every supported evidence level: among the Tier B genes, a phosphomannomutase/phosphoglucomutase, a starch-branching enzyme, an ERD6-like sugar transporter, and a serine carboxypeptidase; and among the two cross-context-supported genes, an alpha-xylosidase acting on xyloglucan. We emphasize that this is a qualitative clustering of family-level annotations over small gene sets, assembled from different genes at different evidence levels, and not a formal over-representation result; no enrichment test was performed, and several annotations are provisional. The pattern is nonetheless consistent with the emphasis on sugar metabolism and cell-wall reinforcement in prior comparisons of resistant and susceptible lychee cultivars [5], and with the general role of apoplastic sugar status and wall remodeling in oomycete resistance. The Tier B set also contains a DIR1-like lipid-transfer protein, whose *Arabidopsis* counterpart is a mobile signal in systemic acquired resistance, and a BAG-family chaperone regulator; BAG proteins act as HSP70 co-chaperones within the chaperone networks that shape plant immune responses and immune-associated cell death [10]. The strong negative discovery enrichment of jasmonic acid signaling, although it did not survive external evaluation, is likewise a plausible axis of cultivar difference given the jasmonate-associated promoter elements suggested in the exploratory stage and the known antagonism between jasmonate and salicylate defense programs. We advance all of these as prioritized, testable hypotheses; the tier system exists precisely to prevent us from advancing them as more.

The external evaluation delivers a second structural lesson. The internally robust and externally supported sets did not overlap, but this disjunction carries almost no information on its own: with 16 and 2 genes selected from 206 candidates, the expected overlap under independence is 0.16 and zero overlap occurs with probability 0.85. The informative observations are continuous ones: discovery and external interaction effects were uncorrelated across the 184 measurable candidates, the frozen signature reversed sign between 24 h and 48

h in the external time course, and the same signature tracked infection in opposite directions in leaf and fruit of the transfer cohort. The straightforward reading is that the cultivar-dependent component of the early infection response is substantially tissue- and time-specific, so that a candidate's failure to transport from leaf at 24 h to pericarp at 48 h is expected behavior rather than evidence against the discovery. This has a practical corollary for breeding applications: expression-based resistance markers derived from one tissue and time point should not be assumed portable, and marker development for this pathosystem will require validation in the tissue and developmental window where deployment is intended.

The orthogonal layers contribute mainly disciplined null results, and we consider their inclusion one of the more useful aspects of the resource. The post hoc controlled PlantCARE comparison now isolates the background rule as far as the archived inputs permit. It shows that genomic expression/GC matching changes which class passes—ARE is replaced by ABRE—but does not eliminate the MeJA, TCA, or TC-rich signals when the published foreground totals are held fixed. Background choice is therefore not a sufficient explanation for the failure of the registered JASPAR analysis; the changed foreground, motif representation, scoring, and robustness rule remain consequential. Because the original observed-motif TSV was not archived and most published totals cannot be reproduced from the current canonical promoters, the controlled result is a provenance audit rather than evidence of regulatory function. The annotation and small-RNA outcomes likewise expose reference-ecosystem gaps (no precomputed InterPro architectures and no *Litchi* entries in the frozen miRNA reference) that bound what any computational study of this species can currently claim.

Several limitations qualify all of the above. The deposited designs are small, with three libraries per cell, and the independence of those libraries at the level of source trees, pooling, and extraction is unreported, so our *q*-values are conditional on deposited-library independence. The public metadata also leave the inoculation protocol underspecified, including the inoculum form (mycelia, sporangia, or zoospores), the developmental stage of the sampled leaves, and whether the tissue was wounded before inoculation, so all conclusions are additionally conditional on the depositors' experimental design. All quantification depends on a single reference assembly, and although uniform and observed mapping checks removed the clearest failure modes, residual reference bias cannot be excluded. The primary external cohort is deliberately heterogeneous with respect to the discovery cohort, which makes support conservative and non-support ambiguous; no genuinely independent replication cohort (same tissue, same time, different biological material) exists in the public record for this design.

Unsupported external outcomes additionally conflate limited power, context dependence, and true absence of effect. The simulations quantify the first component: conditional 80% minimum detectable effects were 2.44 log₂FC (5.4-fold) for genome-wide discovery and 2.11 log₂FC (4.3-fold) for candidate-family external evaluation, with still poorer performance among low-expression genes. These conditional curves do not identify which unsupported genes are context-specific or truly null. And the entire study is computational: no candidate, including the Tier B set, has been perturbed experimentally. The natural next steps are quantitative PCR of the Tier B genes in independently infected material of both discovery cultivars, functional tests of the alpha-xylosidase and DIR1-like candidates, and, for any regulatory hypothesis, direct assays such as promoter-reporter or chromatin-binding experiments rather than further motif counting.

4. CONCLUSIONS

By carrying a single pathosystem from exploratory candidate discovery through a prospectively registered genome-wide discovery framework with locked cross-context external evaluation, this study establishes which claims about cultivar-dependent responses of lychee to *P. litchii* the current public data can and cannot support. It can support a set of 206 QC-retained genome-wide statistical interaction candidates, of which 19 are significant under both statistical frameworks, a stable core of 16 internally robust genes, 12 Tier B candidates whose annotations include cell-wall, sugar-metabolism, lipid-signaling, and RNA-metabolism functions, and 2 genes with cross-context support in an independent tissue and cultivar contrast. It cannot support the original 18 exploratory candidates as validated markers, any pathway- or signature-level claim of context-free transport, or any regulatory-element claim under matched-background testing. All frozen results, including nulls, contradictions, and retirements, are released with the analysis code and checksummed source data, so that the next stage of experimental validation can begin from an honest baseline.

5. MATERIALS AND METHODS

5.1 Study design, registration, and evidence vocabulary

The confirmatory stage was governed by a time-stamped protocol, with amendments logged, distributed with the accompanying repository. The registration chronology is as follows. The exploratory stage was completed and publicly documented first [9], so the discovery dataset had already been analyzed, under different models, before registration. Protocol version 1.0.0 was frozen on 18 August 2026, with its SHA-256 hash (ef668486...) recorded alongside the protocol text, before any confirmatory expression outcome was computed. The amendment

log records 20 outcome-blind amendments (clarifications, implementation corrections, and optimizations, all time-stamped before any discovery or external expression outcome was viewed) and 3 post-outcome amendments confined to technical quality control and manifest verification, each marked as such. The primary external study had been published previously [5], and its headline biological conclusions were known to the analyst before registration; dataset roles and all thresholds were nevertheless fixed before any external count matrix was processed. Because all cohorts are public, the staged external unlock is a workflow-enforced computational safeguard, not blinding in the experimental sense. The protocol fixed dataset roles (Table 1), statistical models, discovery and evaluation thresholds, conditional gates, empty-set behavior, and the vocabulary used throughout this paper: **discovery** refers only to the locked genome-wide interaction analysis in PRJNA830488; **robustness** refers only to predeclared sensitivity analyses within that cohort; **cross-context support** denotes a passing frozen external test in a cohort that differs in tissue, comparator, or design, and is never described as replication; and **orthogonal** evidence classes (annotation, motifs, small RNA, literature) are reported separately from expression support and cannot substitute for it. No external outcome could modify frozen discovery membership, signature weights, or thresholds.

5.2 Exploratory-stage analyses (Stage 1)

The exploratory stage, reported in full elsewhere [9], analyzed the 12 GSE201243 libraries with FastQC v0.11.9 quality control, Trimmomatic v0.39 trimming, TopHat v2.0.12/Bowtie2 v2.2.3 alignment to the SCAU_Lch_v2.0 assembly (GCA_019925255.1), HTSeq v2.0.2 union-mode counting, and DESeq2 v1.42.0 per-cultivar Wald contrasts (design ~ treatment) with Benjamini–Hochberg adjustment at $P < 0.05$ [11–16]. Candidate prioritization combined significance, fold-change magnitude, and annotation plausibility; structural, conserved-motif, and promoter analyses used TBtools-II v2.096, MEME v5.5.5, and PlantCARE [17–19]; and the promoter-background comparison tested observed **cis**-element counts in the 18 candidate promoters against 100,000 randomized sets of eighteen 2-kb sequences with lychee-like GC content, scoring exact motifs and reverse complements, with one-sided empirical P -values adjusted by Benjamini–Hochberg [20]. The selected-gene interaction contrast was computed as $\log_2\text{FC}(\text{Yurong1, infected versus mock}) - \log_2\text{FC}(\text{Guiwei, infected versus mock})$ over the prioritized genes and was explicitly labeled a prioritization effect size rather than a genome-wide test.

5.3 Data provenance and eligibility

National Center for Biotechnology Information and Gene Expression Omnibus metadata for all five cohorts were reconciled into a biological-unit registry (Supplementary S1). The original

repository preserved accession decisions but not the historical query log, so search coverage was reconstructed retrospectively on 27 August 2026 and is not represented as prospective provenance. A broad NCBI GEO series query returned six records and a corresponding BioProject query returned seven; a narrow SRA cross-check returned 42 experiment-level records. After collapsing the GSE222652 super-series, excluding PRJNA1157370 because it used **Colletotrichum gloeosporioides**, and excluding PRJNA268587/GSE63658 because it studied fruit senescence/storage without **P. litchii** inoculation, the five prespecified cohorts remained (Supplementary S16a–b). Because source-tree, pooling, harvest, and extraction independence were not reported for the discovery libraries, all inferential **q**-values are conditional on deposited-library independence. The primary model required at least three included libraries in every cultivar-treatment cell.

5.4 Confirmatory preprocessing and quality gates

Host (SCAU_Lch_v2.0) and pathogen references were combined with prefixed contig names to absorb pathogen-derived reads. Paired reads were checked with FastQC, trimmed with fastp [21], aligned with STAR two-pass mapping [22], counted at the gene level with featureCounts [23], and independently quantified with Salmon [24]. Libraries were excluded if fewer than 10 million read pairs survived trimming or fewer than 40% of reads aligned uniquely; all 12 discovery libraries passed. GenMap-based exon-uniqueness scores [25] and candidate-level observed-mapping checks were fixed before discovery and used for the uniform mapping and observed mapping-sensitivity gates.

5.5 Genome-wide discovery model

DESeq2 [16] fitted gene counts with the design $\sim$ cultivar + treatment + cultivar:treatment (reference levels Guiwei, mock). The primary coefficient, cultivarYurong1:treatmentinfected, estimates the difference in infected-minus-mock response between Yurong1 and Guiwei. Genes required counts of at least 10 in at least three libraries; statistical candidates required genome-wide Benjamini–Hochberg $*q* < 0.05$ and $|\log_2FC| \geq \log_2(1.5)$ [20]. The registered rule tests the point null of zero interaction and applies the effect threshold to the observed estimate. Post hoc sensitivity used DESeq2 with ``lfcThreshold = log2(1.5)`` and ``altHypothesis = greaterAbs``, plus apegglm false-sign-or-small **s**-values for the same threshold; neither analysis modified frozen membership. To align the legacy audit with its original estimand, separate post hoc DESeq2 models (``~ treatment``) were fitted within Guiwei and Yurong1, using the same count matrix, 19,445-gene universe, and genome-wide Benjamini–Hochberg adjustment within each cultivar. Unshrunk maximum-likelihood estimates are reported throughout; apegglm-shrunk interaction estimates [26] were used only to define the signed signature weights over the 206

frozen genes. Mappability and gene-model metrics were computed for all 19,445 tested genes; as a sensitivity check, recomputing the Benjamini–Hochberg adjustment over only the 13,602 QC-eligible genes rediscovered all 206 frozen candidates plus 12 additional genes, confirming that applying quality control after significance testing was conservative here.

5.6 Pathway and transcript-usage discovery

Plant Reactome pathway memberships [27] were transported to lychee through one-to-one reciprocal-best DIAMOND protein matches to *Oryza** [28], and pathway discovery applied the frozen genome-wide interaction statistic to this fixed mapping with camera, roast, and fgsea testing plus leading-edge-deletion and expression/length-matched null gates for robustness [29–31]. Differential transcript usage used Salmon transcript abundances analyzed with DRIMSeq and DEXSeq under stageR stage-wise error control [32–34]; the conditional inferential gate for interpretable two-level error control was registered in advance, and no substitute analysis was permitted had it failed.

5.7 Internal robustness gates

Each frozen gene was retested under Salmon-based gene-level quantification, edgeR quasi-likelihood interaction testing [35], a counts-per-million expression filter, and all leave-one-library-out refits, with predeclared criteria combined conjunctively; no additive scoring was used. The Salmon gate required direction agreement and an absolute effect difference of at most 0.5 log₂ units; the edgeR analysis used the identical design matrix and interaction coefficient with TMM normalization, robust dispersion estimation (estimateDisp and glmQLFit with robust settings), and glmQLFTest, with Benjamini–Hochberg adjustment computed genome-wide over all 19,445 genes and a gate of genome-wide $q^* < 0.10$ plus direction agreement; the expression-filter gate refit the primary model after a counts-per-million filter; and the leave-one-library-out gate required direction agreement in all 12 refits and $q^* < 0.10$ in at least 10 of 12. The observed mapping-sensitivity gate removed candidates whose effect depended on ambiguously mapped reads. Frozen pathways required agreement across camera, roast, leading-edge deletion, and the matched-null comparison.

5.8 External frozen evaluation

PRJNA450886 was analyzed with the design $\sim \text{cultivar} \times \text{treatment} \times \text{time}$, with time encoded as a categorical factor and 24 h as the reference level, so that the cultivar-by-treatment coefficient estimates the 24-h interaction and the three-way terms give the 6-h and 48-h contrasts; the primary estimand was the resistant-minus-susceptible cultivar-by-infection interaction at 24 h, with 6-h and 48-h contrasts secondary. Frozen candidate tests used Benjamini–Hochberg adjustment across the full candidate-by-contrast family, an absolute effect

threshold of $\log_2(1.5)$, confidence intervals excluding zero, and discovery-direction agreement; opposite-direction passes were recorded as contradictory. PRJNA922966 was analyzed with ~ tissue + treatment + tissue:treatment for generic-transfer tests, and PRJNA1090613 outcomes were confined to exploratory status by protocol. The frozen signed signature was evaluated as a library-level score with the same family-wise adjustment.

5.9 Annotation, motif, small-RNA, and literature classes

Representative proteins were chosen by longest coding sequence with lexicographic tie-breaking. Reviewed Swiss-Prot (release 2026_02) DIAMOND matches [36], targeted InterPro/Pfam domain assignments [37], and one-to-one *Oryza* orthology formed three separate annotation classes; the high-confidence label required at least two classes, at least 70% supported coverage, and no architecture conflict. Promoter-motif analysis tested 927 JASPAR plant position-weight matrices [38] in strand-aware 1-kb and 2-kb promoter windows against 100 expression- and GC-matched background sets, with AME as the primary test, FIMO as sensitivity [39,40], robustness requiring passes in at least 80 of 100 backgrounds in both windows, and cognate transcription-factor expression required for attribution. The PmiREN bulk archive [41] was audited before small-RNA analysis; because it contained no exact *Litchi* entries, the class was frozen as not testable and no study-derived reference was substituted. Published studies reusing any analyzed accession were registered as prior interpretation. For the post hoc controlled PlantCARE comparison, published observed totals across the 18 legacy promoters were fixed and tested against (i) 100,000 simulated sets of eighteen 2-kb sequences at 34% GC and (ii) 100 genomic sets of eighteen promoters matched by mean-expression and promoter-GC quintiles with the same seeded minimum-distance rule used by the registered motif pipeline. The six reported motifs and one standard TCA variant, together with reverse complements, were counted and one-sided empirical *P*-values were adjusted across seven classes. The missing archived observed-motif TSV and canonical-promoter recount discrepancies were retained as explicit provenance limitations (Supplementary S15).

5.10 Simulation-based power analysis

Negative-binomial parametric simulations used DESeq2 maximum-a-posteriori gene dispersions and fitted sample means from the discovery counts and from the 24-h subset of PRJNA450886. Fitted cultivar and treatment main effects and size factors were retained, background-gene interaction coefficients were set to zero, and equal numbers of positive and negative effects were injected at 12 absolute $\log_2\text{FC}$ values (0, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 3.5, 4). Each value had 100 simulations. Discovery injected 262 targets and applied Benjamini–Hochberg correction over 19,445 genes plus the registered effect filter. External

simulation injected 2 targets among the 177 frozen genes passing the simulation count and dispersion validity filters and additionally required the prespecified sign and a 95% interval excluding zero; the observed external table contained 184 measurable genes, seven of which were not fit-eligible for the simulation. Dispersions were treated as known during simulated Wald refits, making the curves conditional parametric power rather than a claim about population-level biological replication. Eighty-percent minimum detectable effects were linearly interpolated between grid points, with expression-quartile curves and Wilson intervals reported in Figure S2.

5.11 Tier rules, software versions, and reproducibility

Tier A required discovery, complete internal robustness, primary cross-context support, clean mapping and annotation, and at least one attributable orthogonal class; Tier B required discovery and complete internal robustness with partial or non-testable primary external evidence; Tier C was reserved for a robust, externally supported pathway. Contradictory external direction or mapping/annotation failure retired an entity, and an empty Tier A was an allowed outcome. All stages ran under Snakemake 9.25.2 workflows [42] enforcing staged discovery, a controlled external unlock, verified cleanup, and SHA-256 manifests over inputs and outputs. Confirmatory executables were FastQC 0.12.1, fastp 1.3.6, STAR 2.7.10b, featureCounts/Subread 2.1.1, Salmon 2.5.1, MEME Suite/AME/FIMO 5.5.9, and DIAMOND 2.2.5; GenMap 1.3.0 is recoverable from the pinned specification although its executable is absent from the reconstructed environment. Statistical work used R 4.3.3/Bioconductor 3.18 with DESeq2 1.42.0, apegglm 1.24.0, edgeR 4.0.16, DRIMSeq 1.30.0, DEXSeq 1.48.0, stageR 1.24.0, and BiocParallel 1.36.0 (Supplementary S17 records the evidence status of every version). The resolved environment, synthetic fixtures, per-figure source data, and amendment log are distributed with the repository. All six main figures are regenerated from frozen result tables by a script that recomputes and asserts every headline count.

DECLARATIONS

****Funding:**** This work received no external funding. ****Conflicts of interest:**** The author declares no conflicts of interest. ****Author contributions:**** E.Z. conceived the study, performed all analyses, and wrote the manuscript.

DATA AND CODE AVAILABILITY

All analyzed data are public: PRJNA830488/GSE201243 (discovery), PRJNA450886 (primary external), PRJNA922966/GSE222651 (generic transfer), PRJNA922965/GSE222650 (small RNA), and PRJNA1090613/GSE262200 (exploratory). The registered protocol, amendment log, complete result tables (all discovery statistics, robustness manifests, external

754 tests, pathway/signature tests, transcript-usage results, annotation and orthogonal evidence,
755 and the contradictory-results file), and the workflow code accompany this repository; a
756 permanently archived release with a DOI, matching the submitted manuscript, will accompany
757 the revised submission. Exploratory-stage scripts and supplementary tables are available with
758 the original report [9].

759 **SUPPLEMENTARY INFORMATION**

760 Supplementary tables S1–S18 provide the biological-unit registry, per-library quality control,
761 all 19,445 discovery statistics (including composite-null and false-sign-or-small columns),
762 robustness manifests, all external frozen tests, pathway/signature and transcript-usage results,
763 annotation and orthology evidence, the small-RNA gate, all registered motif-background tests,
764 the evidence registry, script/environment inventory, amendment log, within-cultivar legacy audit
765 (S14), controlled PlantCARE background comparison (S15), reconstructed search queries and
766 eligibility decisions (S16a–b), exact software versions (S17), and power minimum-detectable
767 effects (S18). Figure S1 shows replicate-level normalized counts, Figure S2 the conditional
768 power curves, and Figure S3 the quarantined exploratory signature estimate. Every main and
769 supplementary analytical figure has tab-separated source data where applicable.

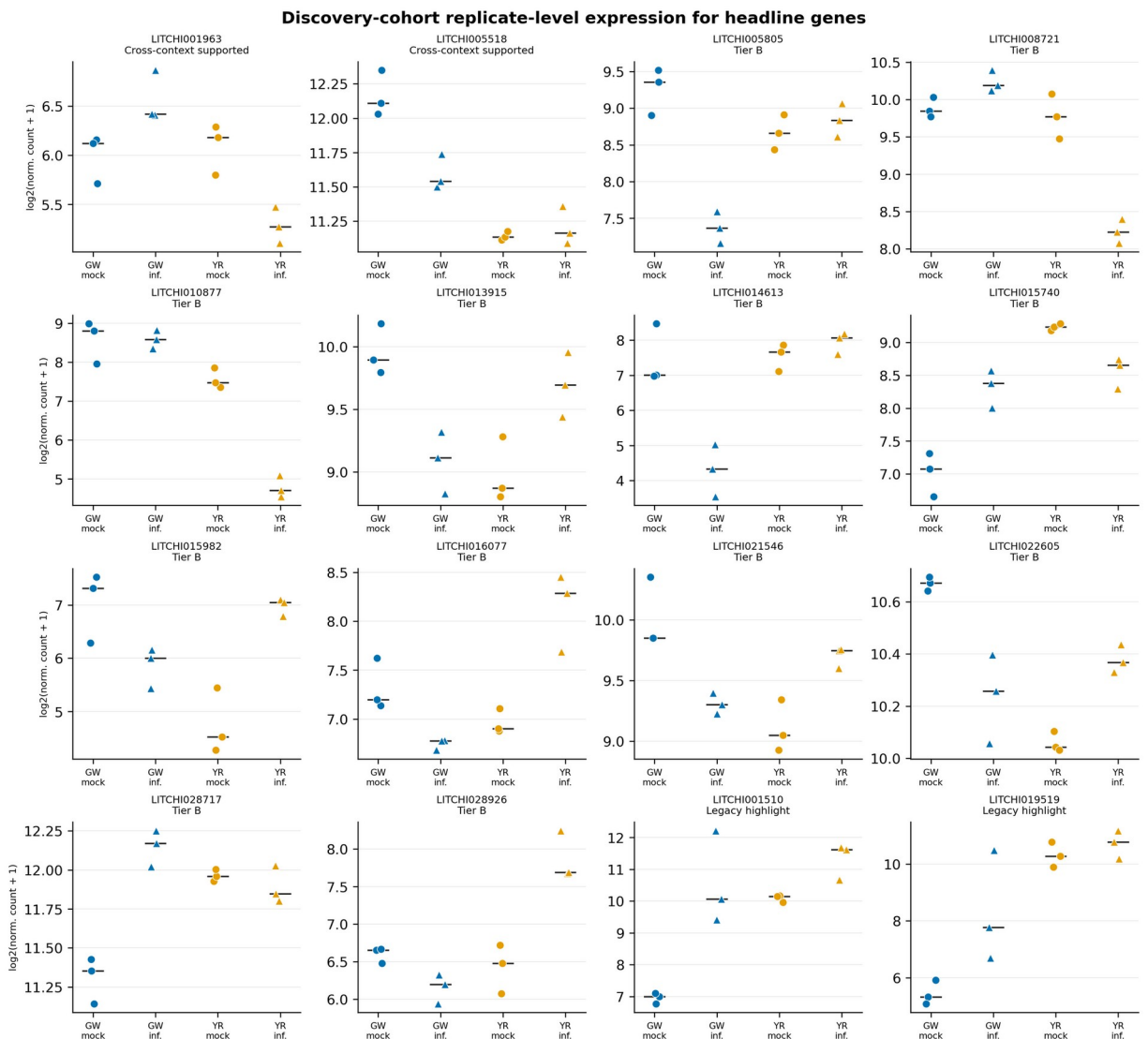
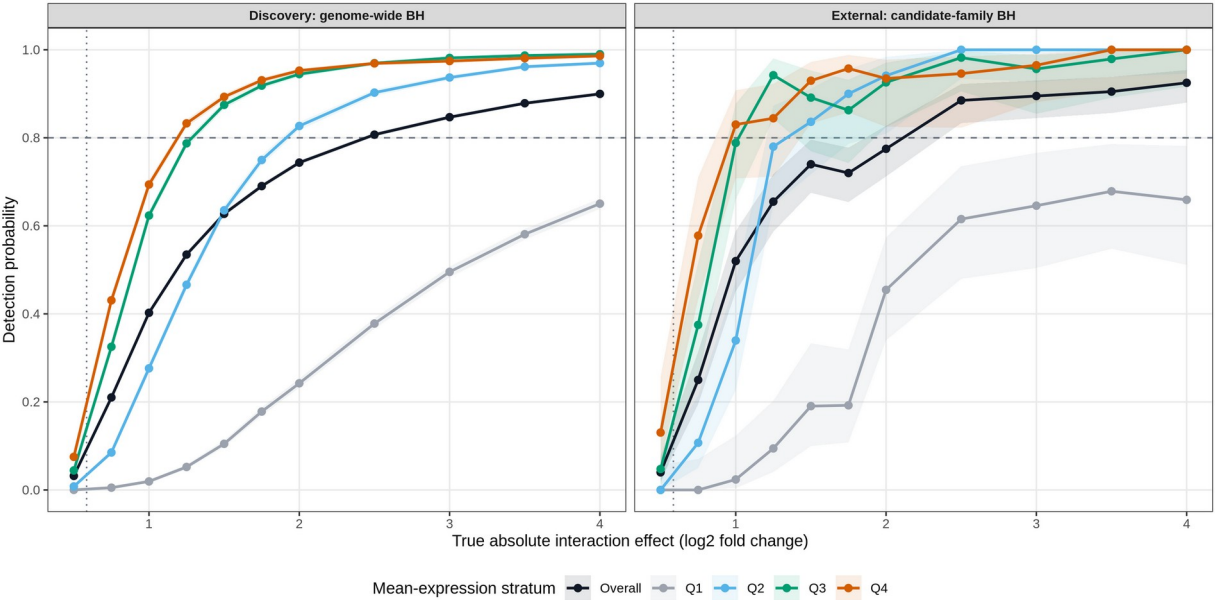


Figure S1. Replicate-level normalized counts for the two cross-context-supported genes, twelve Tier B genes, and two legacy highlights. Points are individual deposited libraries; bars show cultivar-treatment medians.



Dashed line: 80% power. Dotted line: registered $|\log_2FC|$ threshold. Parametric power is conditional on fitted means, dispersions, deposited-library independence, and the simulated non-null prevalence.

Figure S2. Parametric detection probability for cultivar-by-infection effects under genome-wide discovery and candidate-family external adjustment. Curves show the overall result and mean-expression quartiles; the dashed line marks 80% power.

S3 Exploratory signature estimate (not confirmatory)

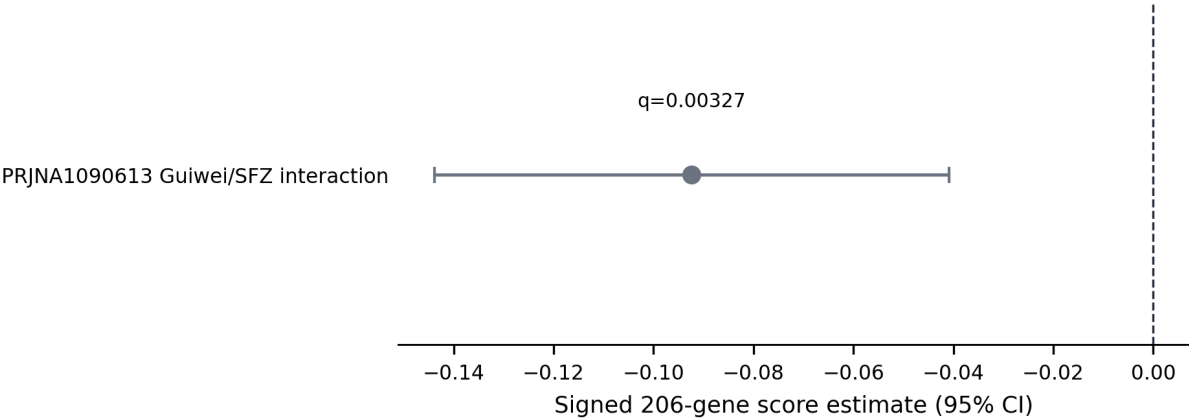


Figure S3. Quarantined PRJNA1090613 signed-signature estimate, retained as exploratory rather than confirmatory evidence.

Electronic supplementary data

The complete machine-readable tables are distributed with the repository and are not embedded in this DOCX because several contain thousands to hundreds of thousands of rows. Every item is checksum-covered by the release bundle.

Electronic supplementary data inventory.

Artifact	Contents
results/tables/Table1_dataset_roles_eligibility.tsv	Dataset roles, estimands, eligibility, and locked limitations. Rows: 6.
results/tables/Table2_frozen_discovery.tsv	All frozen gene and pathway discoveries. Rows: 268.
results/tables/Table3_internal_robustness.tsv	Gene and pathway internal-robustness outcomes. Rows: 212.

Artifact	Contents
results/tables/Table4_external_evaluation.tsv	All external frozen gene and pathway evaluations. Rows: 1,460.
results/tables/Table5_orthogonal_final_status.tsv	Orthogonal evidence and deterministic final status. Rows: 268.
results/supplement/S10_motif_background_tests.tsv	Motif/background tests and sensitivity results. Rows: 752,724.
results/supplement/S11_evidence_registry.tsv	Accession-aware published-evidence registry. Rows: 5.
results/supplement/S12_scripts_environments_commands.tsv	Scripts, environments, and command inventory. Rows: 62.
results/supplement/S13_amendment_deviation_log.tsv	Protocol amendments and deviations. Rows: 24.
results/supplement/S14_legacy_within_cultivar_audit.tsv	Electronic evidence table Rows: 36.
results/supplement/S15_controlled_motif_background.tsv	Electronic evidence table Rows: 21.
results/supplement/S16a_dataset_search_queries.tsv	Electronic evidence table Rows: 3.
results/supplement/S16b_dataset_eligibility.tsv	Electronic evidence table Rows: 8.
results/supplement/S17_exact_tool_versions.tsv	Electronic evidence table Rows: 19.
results/supplement/S18_power_simulation_mde.tsv	Electronic evidence table Rows: 10.
results/supplement/S1_biological_unit_registry.tsv	Biological-unit and accession registry. Rows: 84.
results/supplement/S2_per_library_QC.tsv	Per-library discovery QC. Rows: 12.
results/supplement/S3_all_discovery_statistics.tsv	Genome-wide discovery statistics. Rows: 19,445.
results/supplement/S4_robustness_file_manifest.tsv	Internal robustness file manifest. Rows: 20.
results/supplement/S5_all_external_frozen_tests.tsv	All external frozen tests. Rows: 2,135.
results/supplement/S6_pathway_signature_tests.tsv	Pathway and signature tests. Rows: 25.
results/supplement/S7_DTU_results.tsv	Conditional differential-transcript-usage results. Rows: 15,790.
results/supplement/S8_annotation_orthology.tsv	Candidate annotation and orthology evidence. Rows: 206.
results/supplement/S9_small_RNA_results.tsv	Small-RNA reference-gate outcome. Rows: 1.

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TABLES

Table 1. Locked dataset roles and eligibility (condensed from electronic Table 1).

Accession	Design	Role	Eligibility
PRJNA830488	Guiwei/Yurong1 leaf; mock/infected; 24 h; 3 libraries/cell	discovery	conditional_inferential
PRJNA450886	Guiwei/Heiye fruit pericarp; mock/infected; 6/24/48 h; 3 libraries/cell	primary_cross_context	cross_context
PRJNA922966	Feizixiao leaf/fruit; mock/infected; 24 h; 3 libraries/cell	generic_infection_transfer	cross_context
PRJNA922965	Feizixiao leaf/fruit small RNA; mock/infected; 24 h; 3 libraries/cell	orthogonal_modality	orthogonal
PRJNA1090613	Guiwei/SFZ leaf; mock/infected; 3 libraries/cell	exploratory	exploratory

Table 2. Key validated computational outcomes.

Metric	Value
included discovery libraries	12
expressed genes tested	19445
statistical gene candidates	262
frozen gene discoveries	206
internally robust genes	16
frozen pathways	6
internally robust pathways	1
primary external cross context genes	2
primary external contradictory genes	5
primary external cross context pathways	0
frozen DTU rows	225

Metric	Value
high confidence annotations	0
robust discovery motifs	0
externally transported motifs	0
Tier A	0
Tier B	12
Tier C	0
Retired	65
contradictory or failed rows	65

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895 Table 3. Deterministic final tier distribution.

Tier/status	Count
Tier A	0
Tier B	12
Tier C	0
Exploratory	191
Retired	65

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